INTR 5100 Traffic Flow Theory

Lab Session I Guidelines

Car-Following Model Calibration

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1 Objective

Lab Session I focuses on human car-following data collection, microscopic car-following model calibration, and model validation. The experiment has three stages:

1. collect human car-following trajectory data;
2. calibrate an Intelligent Driver Model (IDM) from the collected data; and
3. validate the calibrated IDM in a separate physical experiment.

The vehicle configuration during human-driving data collection is

$$\text{Leading vehicle} \longrightarrow \text{Human-driven following vehicle}. \tag{1}$$

The leading vehicle follows a predefined speed-command pattern, while one student controls the physical following vehicle through the driving simulator.

2 Preparation During the Week Before the Lab

Before coming to Lab Session I, each group should:

- review the IDM formulation and parameter meanings;
- understand the trajectory-file structure described at the end of this guideline; and
- prepare a working calibration environment for estimating the five IDM parameters (v_0, T, s_0, a, b) from the experimental trajectory data.

The choice of calibration method and software tool is open. For the IDM, let s denote the bumper-to-bumper spacing, v the following-vehicle speed, v_{L} the leading-vehicle speed, and

$$\Delta v = v - v_{\mathrm{L}} \tag{2}$$

the approaching rate. The IDM acceleration is

$$a_{\mathrm{IDM}}(s, v, \Delta v) = a \left[1 - \left(\frac{v}{v_0} \right)^{\delta} - \left(\frac{s^*}{s} \right)^2 \right], \tag{3}$$

where

$$s^* = s_0 + \max \left\{ 0, vT + \frac{v \Delta v}{2\sqrt{ab}} \right\}. \tag{4}$$

The corresponding car-following dynamics are

$$\dot{s} = v_{\mathrm{L}} - v, \qquad \dot{v} = a_{\mathrm{IDM}}(s, v, \Delta v). \tag{5}$$

The parameters to be calibrated are

$$\boldsymbol{\theta} = [v_0, T, s_0, a, b]^{\top}, \tag{6}$$

while the acceleration exponent is fixed at

$$\delta = 4. \tag{7}$$

Table 1: IDM parameters to be calibrated.

Parameter	Description
v_0	Desired speed
T	Desired time gap
s_0	Minimum gap
a	Maximum acceleration
b	Desired deceleration

3 During the Lab Session

3.1 Human-Driving Data Collection

The laboratory session begins with the two-vehicle human-driving experiment.

1. One student operates the driving simulator and controls the physical following vehicle.
2. The following vehicle follows the leading vehicle for one complete route.
3. The driver remains behind the leading vehicle and maintains safe and natural car-following behavior.
4. The leading- and following-vehicle trajectories are recorded.
5. The human-driving dataset is provided to the group after the run.

The complete route is recorded. The calibration uses the continuous straight-road portions of the leading- and following-vehicle trajectories.

3.2 IDM Calibration

Each group then applies its prepared calibration method to the collected human-driving data and estimates

$$(v_0, T, s_0, a, b). \tag{8}$$

The calibration method should include an appropriate fitting criterion and a numerical procedure for estimating the five parameters. The calibration result should also be evaluated using a quantitative measure selected by the group. One complete IDM parameter set must be obtained during the lab session and submitted for the validation experiment.

3.3 Model Validation

The calibrated IDM is then configured on the physical following vehicle. The vehicle configuration becomes

$$\text{Leading vehicle} \longrightarrow \text{IDM following vehicle}. \tag{9}$$

A separate validation run is conducted using the calibrated parameter set. During the validation run, the calibrated IDM controls the following vehicle when both vehicles are in straight-road regions. In curved-road regions, the following vehicle receives a fixed speed command. A new trajectory dataset is recorded and provided to the group for subsequent validation analysis.

4 After the Lab Session

After the laboratory session, each group should complete the calibration and validation analysis for the final project report. The analysis should include:

- trajectory preprocessing;
- the selected IDM calibration method;
- calibrated IDM parameters and their interpretation;
- calibration and validation performance;
- comparison between measured and model-generated trajectories; and
- evaluation of the calibrated IDM's accuracy and limitations.

5 Data Format

The human-driving and validation runs use the same general trajectory-file structure. The main trajectory files are:

- `straight_trajectory.csv`: combined straight-road records of the leading and following vehicles;
- `leader_straight_trajectory.csv`: straight-road records of the leading vehicle; and
- `follower_straight_trajectory.csv`: straight-road records of the following vehicle.

Each run also contains `leader_events.csv`, `commands.csv`, and `metadata.json`. The `commands.csv` file is empty for the manually driven follower during the human-driving run and contains the automatic following-vehicle control records during the validation run. The main columns used in the trajectory analysis are summarized in Table 2.

Table 2: Main trajectory variables.

Column	Meaning
`t_from_start`	Recording time relative to the beginning of the experimental run
`role`	Vehicle role
`x`, `y`	Measured two-dimensional vehicle position
`heading`	Vehicle heading
`velocity`	Vehicle speed
`s`	Longitudinal route coordinate obtained by projecting the vehicle position onto the predefined route
`zone_type`	Route-region classification: straight, approach, corner, or exit

A simplified excerpt of `straight_trajectory.csv` is shown below. The column names follow the experimental trajectory files.

```
t_from_start,role,x,y,heading,velocity,s,zone_type
0.106820,leader,1975.217298,480.000000,0.000000,50.000000,1605.008113,straight
0.107269,follower,1373.933496,484.874337,-7.666804,49.999653,1003.008121,straight
0.213979,leader,1981.365441,480.000000,0.000000,50.000000,1611.156256,straight
0.214815,follower,1380.026987,484.054060,-7.666804,50.005167,1009.156574,straight
```

The variable `s` stored in the trajectory files is the longitudinal route coordinate. Let d denote the forward route distance from the following vehicle to the leading vehicle. The bumper-to-bumper spacing is calculated as

$$s = d - L_{\text{veh}}, \tag{10}$$

where

$$L_{\text{veh}} = 250. \tag{11}$$

Since the leading and following vehicles are recorded independently, their `t_from_start` values are not necessarily identical. The two trajectories should therefore be synchronized within each retained continuous straight-road segment according to the selected calibration or validation procedure.